

MUHAMMAD MUZAMMIL IRSHAD

Multan, Punjab

 [E-mail](#)

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 [GitHub](#)

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 [YouTube](#)

SUMMARY

Computer Science undergraduate working at the intersection of machine learning research and applied AI. Research Assistant at RAISE Lab focused on adaptive learning under concept drift and deep learning for real-world classification tasks; also builds AI agents and automation pipelines with n8n and LangChain, and teaches Python and Machine Learning on YouTube.

EXPERIENCE

Researcher

Research in AI for Science and Engineering (RAISE) Lab

- Researching incremental learning for insider threat detection under concept drift, and deep learning models to distinguish harmful cyberbullying from non-harmful interactions.

Ongoing

Remote

Instructor

YouTube – Auto AI Academy

- Teaching Python, data analysis, and machine learning through tutorial series, turning complex concepts into practical, hands-on lessons.

Ongoing

Remote

AI & Automation Intern

IFFA Tech

- Built a WhatsApp-based restaurant booking and ordering assistant using n8n, automating customer interactions end-to-end.

Completed

Lahore, Pakistan

EDUCATION

University of Education, Lahore (Vehari Campus)

BS CS (undergraduate) - CGPA - 3.8

07, 2023 — 07, 2027

Vehari, Pakistan

Prime College, Tibba Sulatan Pur

Intermediate - Percentage - 78.55%

08, 2021 — 04, 2023

Tibba Sultan Pur, Pakistan

TECHNICAL SKILLS

Programming Languages: Python, C, C++, HTML, CSS, JavaScript

AI/ML: Supervised Learning, Unsupervised Learning, CNN, Transfer Learning, YOLO, Explainable AI (Grad-CAM)

Libraries/Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, Flask

AI Agents / Automation: n8n, LangChain, LangGraph, Hugging Face

Tools: VS Code, Jupyter Notebook, Git, GitHub

PROJECTS

Explainable Wrist Fracture Detector — [Live Demo](#)  — [GitHub](#) 

- Fine-tuned a pretrained DenseNet121 model for wrist X-ray fracture classification from medical imaging data.
- Improved model interpretability using Grad-CAM visualizations for clinically relevant predictions.
- Deployed as a web-based diagnostic demo on Hugging Face Spaces.

Personal Protective Equipment Detector — [Live Demo](#)  — [GitHub](#) 

- Fine-tuned a pretrained DenseNet121 model to detect safety equipment (helmets, vests, gloves) in workplace imagery.
- Evaluated model performance using standard metrics, achieving strong predictive results.
- Deployed on Hugging Face Spaces for real-time, public inference.

Social Media Content Auto-Poster

- Designed an n8n automation that sources, formats, and schedules content for posting via the Buffer API.
- Used a Groq LLM agent to auto-generate post titles and descriptions from source content.
- Added WhatsApp status notifications and Google Sheets logging for pipeline visibility.

AI Restaurant Booking & Ordering Assistant

- Built a WhatsApp-based conversational AI agent for table booking and food ordering using n8n and a Groq LLM agent.
- Implemented tool-calling for menu lookup, order placement, and booking creation, modification, and cancellation.
- Added per-phone-number session memory for context-aware, multi-turn conversations.